

Introduction to Natural Language Processing

Data Processing

Master in Telecommunication Engineering

Jerónimo Arenas-García, Lorena Calvo-Bartolomé, Jesús Cid-Sueiro

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Department of Signal Theory and Communications
Universidad Carlos III de Madrid

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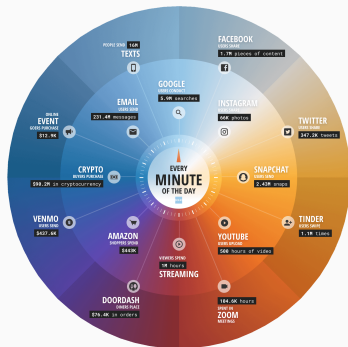
2. What is NLP?

3. Evolution of NLP

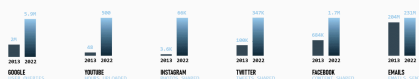
4. Some NLP applications

5. Upcoming topics in this NLP block

Motivation I



DATA NEVER SLEEPS 1.0 VS. 10.0



GLOBAL INTERNET POPULATION GROWTH



As of April 2022, the internet reaches 63% of the world's population, representing roughly 5 billion people. Of this total, 4.65 billion - over 93 percent - were social media users. According to Statista, the total amount of data predicted to be created, captured, copied and consumed globally in 2022 is 97 zettabytes, a number projected to grow to 181 zettabytes by 2025.

To succeed in an increasingly digital world where the volume of data created keeps accelerating, businesses need the right tools to put that data to work right where work gets done. Domo gives you the power to rapidly unlock value from all your data, regardless of where it lives, and drive actions across your organization that will improve business outcomes. Every click, swipe, share, or like tells a story, and Domo helps you do something powerful with it.

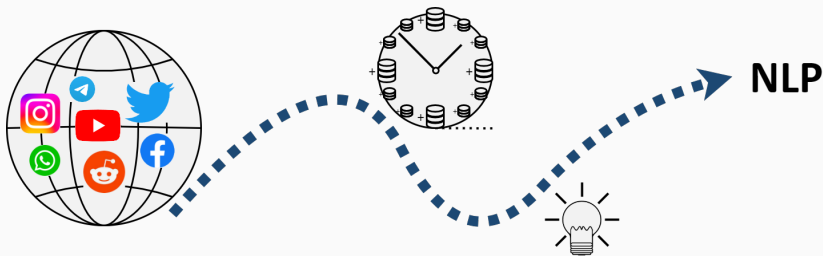
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SOURCES

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<https://www.domo.com/data-never-sleeps>



Large amounts of unstructured text data are generated every second

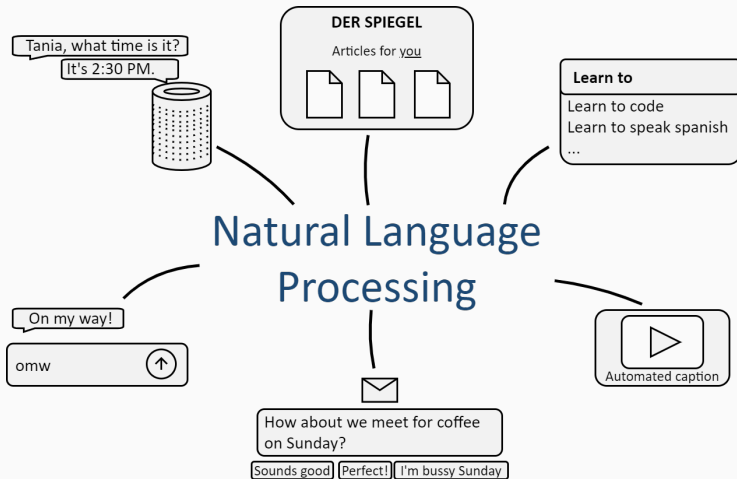
- » **We can no longer use the common approach to understand the text and this is where NLP comes in**

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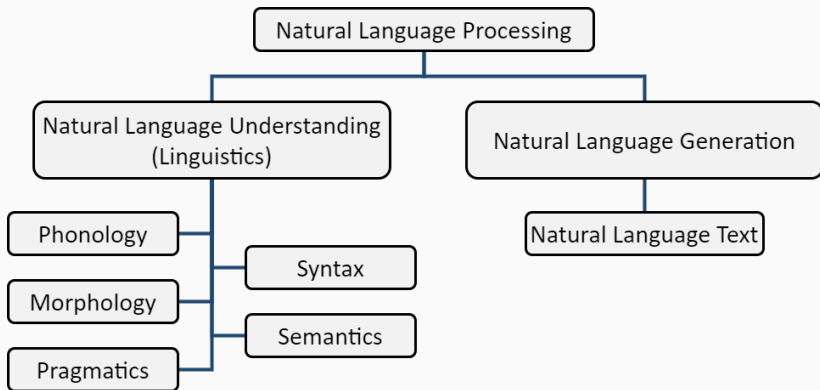
What is NLP?

AI's discipline concerned with giving computers the ability to understand written and spoken human languages in the same manner humans do



Modified from 

Components of NLP

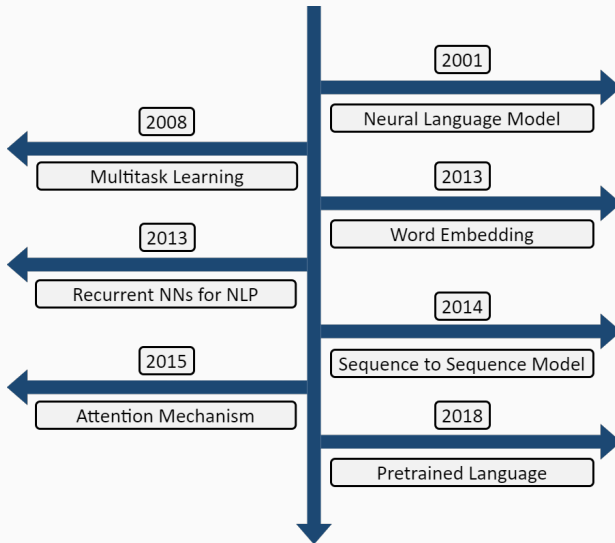


Modified from: Natural language processing: state of the art, current trends and challenges 

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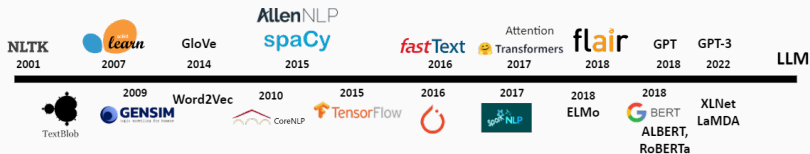
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A walk-through of recent developments in NLP



Natural language processing: state of the art, current trends and challenges 📖

Recent NLP Developments and Tools Progression



Evolution and Progression of NLP Tools in Python 🐍

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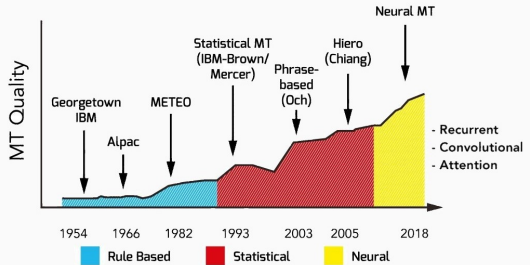
Some NLP applications

Treffen wir uns am Sonntag
zum Kaffee?

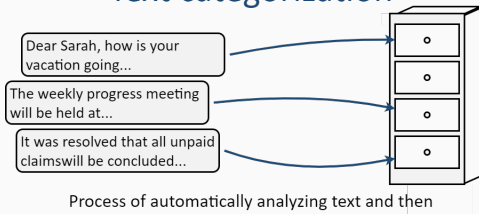
Machine translation

¿Nos vemos el domingo para
tomar un café?

The application of computers to
the task of translating texts from
one natural language to another



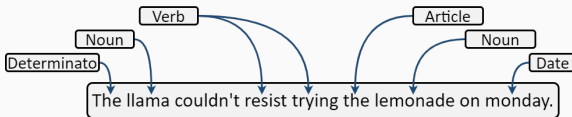
Text categorization



Process of automatically analyzing text and then assigning a set of pre-defined tags or categories based on its content.

- Spam Filtering
- Sentiment Analysis
- Alert detection
- Automating customer walkthroughs in support systems
- ...

Information extraction

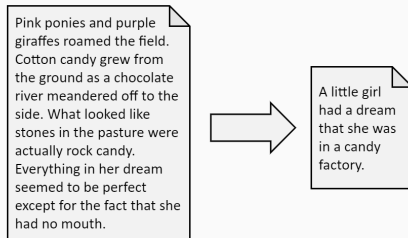


Process of extracting information from unstructured textual sources to enable finding entities (names, places, events, dates, etc.) that can be applied for a variety of purposes, e.g., identify keywords, classifying text items, etc.

Some information extraction techniques:

- POS tagging
- Regular expressions
- NER
- Topic Modeling
- Rule-Based Matching

Summarization



Extractive

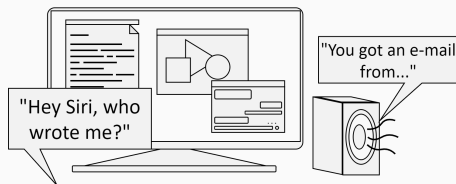
From sentences within the text.

Abstractive

Possible to contain words not explicitly present in the text.

Text summarization is the problem of reducing the number of sentences and words of a document without changing its meaning.

Dialogue Systems



Computer systems intended to converse with a human, employing one or more of text, speech, graphics, haptics, gestures, and other modes for communication on both the input and output channel.



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Upcoming topics in this NLP block



Text Vectorization

One-hot encoding. Bag of Words Representation.
Word Embeddings. Word2Vec. Glove.



Transformers

Introduction to Transformers.
Transfer Learning with Transformers.
Hugging Face.



Semantic Analysis

Semantic Similarities Metrics.
Semantic Information Retrieval.
Graph Visualization & Analysis.